COUNTING PARTIAL HADAMARD MATRICES IN THE NEAR-QUADRATIC REGIME

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Abstract. Let $N_{n,m}$ denote the number of $n \times m$ matrices with entries in $\{\pm 1\}$ whose rows are pairwise orthogonal, and suppose that m is a multiple of four. We prove an explicit, uniform asymptotic formula in a near-quadratic range. Writing

$$A_{n,m} = 2^{nm+(n-1)^2}(2\pi m)^{-n(n-1)/4}, \qquad R_{n,m} = \exp\Big(-\frac{n(n-1)(2n-1)}{24m}\Big),$$

we show that $N_{n,m} = [1 + o(1)]A_{n,m}R_{n,m}$ uniformly whenever $m/(n^2 \log(2\pi m)) \to \infty$. In particular, for every fixed $\eta > 0$ the formula holds uniformly in the regime $m/n^{2+\eta} \to \infty$. The correction factor $R_{n,m}$ is essential below the cubic scale and resums the leading cubic-scale correction. Consequently the infimum of the admissible power-law exponents is at most 2, improving the previous bound 3; validity at the endpoint exponent 2 is not claimed.

The proof proceeds by adjoining rows one at a time. For a fixed k-row partial Hadamard matrix, the number of admissible next rows is a k-dimensional Fourier integral. Exact orthogonality identities give a uniform central expansion for every partial Hadamard matrix. Away from the universal saddle lattice, a scale-adapted truncation and a Bernstein bound show exponential decay for all but an exponentially small set of sign matrices. A quantitative induction then shows that this exceptional set is negligible among partial Hadamard matrices as soon as $m \gg n^2 \log m$. We also give a closed-form error bound valid for every integer pair $n \geq 2$ and $m \in 4\mathbb{N}$.

1. Introduction

A real partial Hadamard matrix is a rectangular sign matrix whose rows are pairwise orthogonal. These objects interpolate between elementary orthogonal sign systems and square Hadamard matrices, and their counting problem is naturally linked to lattice random walks and Fourier inversion. De Launey and Levin introduced a Fourier-analytic framework for this problem and obtained asymptotic counting results for fixed row dimension and, implicitly, in a high-dimensional range [3]. Subsequent work reduced the required separation between the width and the number of rows. In particular, Davis proved a precise expansion at and above the cubic scale $m \asymp n^3$, extending an unpublished estimate of Canfield [1, 2].

The standard leading term is

$$A_{n,m} = 2^{nm+(n-1)^2}(2\pi m)^{-n(n-1)/4}.$$

At the cubic scale this term alone is no longer asymptotically exact. The first correction found in [2] is of order n^3/m. The main point of the present argument is that the whole correction can be resummed into the elementary factor

$$R_{n,m} = \exp\Big(-\frac{n(n-1)(2n-1)}{24m}\Big).$$

Once this factor is retained, the central part of the Fourier integral remains stable down to the near-quadratic scale. The remaining obstacle is not the local expansion but the possibility of additional, highly structured far-field peaks. Such peaks do occur for exceptional partial Hadamard matrices, so a uniform pointwise bound for every matrix is false; see Remark 6.11. Instead, we prove that the exceptional matrices are exponentially rare among all sign matrices, and then compare this rarity with a quantitative lower bound for the density of partial Hadamard matrices.

Our natural asymptotic condition is

$$\frac{m}{n^2\log(2\pi m)} \longrightarrow \infty.$$

This implies the formula in every power-law regime $m/n^{2+\eta} \to \infty$ with fixed $\eta > 0$. It does not settle the endpoint condition $m/n^2 \to \infty$: for example, $m = n^2 \log\log n$ does not force $m/(n^2 \log m) \to \infty$. The endpoint remains outside the reach of the present exceptional-set argument.

Relation to the cubic expansion. If $m = 4t$ and $t = \Theta n^3$ with fixed $\Theta > 0$, then

$$R_{n,4t} \longrightarrow e^{-1/(48\Theta)}.$$

For large Θ this gives $e^{-1/(48\Theta)} = 1 - \frac{1}{48\Theta} + O(\Theta^{-2})$, which agrees with the leading correction in [2]. Thus the present formula is consistent with the known cubic expansion while also predicting the full limiting correction factor.

2. Main results

Throughout the paper, $m = 4t \geq 4$ and

$$L := \log(2\pi m) = \log(8\pi t).$$

For integers $n \geq 2$ and $t \geq 1$, define

$$B_{n,m} := 2^{nm+(n-1)^2}(2\pi m)^{-n(n-1)/4}\exp\Big(-\frac{n(n-1)(2n-1)}{24m}\Big), \qquad (m = 4t), \tag{1}$$

$$\begin{aligned} E(n,t) := \frac{n^4}{4m^2} + 2ne^{-m/(4n)} + 2ne^{2n-m/(86n)} + 3ne^{(n/2)L-m/32} \\ + 2n\exp\Big(2n^2L + 2n\log(200n^2) - \frac{m}{1024}\Big), \end{aligned} \tag{2}$$

$$S(n,t) := \Big(\frac{n^2L}{4} + \frac{n^3}{12m}\Big)\Big(\frac{20000\,n^2L}{m}\Big)^{n^2}, \tag{3}$$

$$\varepsilon(n,t) := \exp\big(E(n,t) + S(n,t)\big) - 1. \tag{4}$$

Theorem 2.1 (Explicit all-parameter estimate)**.** *For every pair of integers $n \geq 2$ and $t \geq 1$,*

$$\Big|\frac{N_{n,4t}}{B_{n,4t}} - 1\Big| \leq \varepsilon(n,t).$$

The closed form in (4) is intentionally conservative. Its purpose is to make the statement valid without side conditions, including parameter pairs for which no $n \times 4t$ partial Hadamard matrix can exist.

Theorem 2.2 (Near-quadratic uniform asymptotic). *Let (n_i, t_i) be any sequence with $n_i \geq 2$, $t_i \geq 1$, and put $m_i = 4t_i$. If*

$$\frac{m_i}{n_i^2 \log(2\pi m_i)} \longrightarrow \infty,$$

then $N_{n_i,m_i}/B_{n_i,m_i} \to 1$. The convergence is uniform in the displayed regime.

Corollary 2.3 (Every exponent strictly larger than two). *For every fixed $u > 2$,*

$$N_{n,4t} = [1 + o(1)]B_{n,4t}$$

uniformly whenever $t/n^u \to \infty$.

If a critical exponent is defined as the infimum of the admissible power exponents, Corollary 2.3 gives an upper bound of 2. This infimal statement must not be confused with validity at the endpoint $u = 2$, which is not proved here. Section 10 makes the infimum statement precise, and Section 12 records what is not proved.

3. Fourier representation and row extension

Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, with total Lebesgue measure 2π. For a $k \times m$ sign matrix H, let $h_1, \ldots, h_k \in \{\pm 1\}^m$ be its rows and $H_1, \ldots, H_m \in \{\pm 1\}^k$ its columns. Let $\mathcal{P}_k$ be the set of k-row partial Hadamard matrices of width m, so that $|\mathcal{P}_k| = N_{k,m}$ and $N_{1,m} = 2^m$.

For $\theta \in \mathbb{T}^k$, define

$$\Phi_H(\theta) := \prod_{C=1}^{m} \cos(\theta \cdot H_C),$$

and let $X(H) := \#\{x \in \{\pm 1\}^m : Hx = 0\}$. Every $(k+1)$-row partial Hadamard matrix is uniquely obtained by adjoining one such row, hence

$$N_{k+1,m} = \sum_{H \in \mathcal{P}_k} X(H). \tag{5}$$

Lemma 3.1 (Fourier representation). *For every $k \times m$ sign matrix H,*

$$X(H) = 2^m (2\pi)^{-k} \int_{\mathbb{T}^k} \Phi_H(\theta)\, d\theta.$$

Proof. For an integer z, $\mathbf{1}_{\{z=0\}} = (2\pi)^{-1} \int_{\mathbb{T}} e^{iz\theta}\, d\theta$. Apply this identity to each coordinate of Hx, sum over $x \in \{\pm 1\}^m$, and factor the resulting product:

$$X(H) = (2\pi)^{-k} \int_{\mathbb{T}^k} \prod_{C=1}^{m} \Big(\sum_{x_C = \pm 1} e^{ix_C(\theta \cdot H_C)} \Big) d\theta = 2^m (2\pi)^{-k} \int_{\mathbb{T}^k} \Phi_H(\theta)\, d\theta. \qquad \square$$

4. The universal saddle lattice

Define

$$\Lambda_k := \Big\{ \tfrac{\pi}{2} c \pmod{2\pi} : c \in \mathbb{Z}^k,\ \textstyle\sum_{j=1}^k c_j \equiv 0 \pmod 2 \Big\} \subset \mathbb{T}^k.$$

We call Λ_k the universal saddle lattice. It is universal because every partial Hadamard integrand is exactly periodic under its translations. It need not contain every point where a particular integrand has modulus one; exceptional matrices can have additional far-field maxima, as shown in Remark 6.11.

Lemma 4.1. *One has $|\Lambda_k| = 2^{2k-1}$, and distinct points of Λ_k are at least $\pi/\sqrt{2}$ apart in $\mathbb{T}^k$.*

Proof. Modulo 2π, take $c \in \{0,1,2,3\}^k$. Exactly half of the 4^k classes have even coordinate sum. For the distance statement, let $c'' \not\equiv 0 \pmod 4$ be the difference of two even-sum representatives. If some coordinate of c'' is congruent to 2 modulo 4, the distance is at least π. Otherwise every coordinate is odd or $\equiv 0$, the number of odd coordinates is positive and even, hence at least two, giving distance at least $\sqrt{2(\pi/2)^2} = \pi/\sqrt{2}$. □

Lemma 4.2 (Exact periodicity)**.** *Let H be a k-row partial Hadamard matrix and suppose $4 \mid m$. Then for every $\omega \in \Lambda_k$ and $\beta \in \mathbb{T}^k$, $\Phi_H(\omega+\beta) = \Phi_H(\beta)$.*

Proof. Write $\omega = (\pi/2)c$ with $\sum_j c_j$ even. For each column C, $\sum_j c_j H_{jC}$ is even, and therefore $\omega \cdot H_C \in \pi\mathbb{Z}$. Thus

$$\cos((\omega+\beta)\cdot H_C) = \epsilon_C \cos(\beta \cdot H_C), \qquad \epsilon_C \in \{\pm 1\}.$$

Let $s_j = \sum_C H_{jC}$ be the jth row sum. Then $\prod_C \epsilon_C = (-1)^{\frac{1}{2}\sum_j c_j s_j}$. It remains to prove $\sum_j c_j s_j \equiv 0 \pmod 4$. Every s_j is even. Moreover, if $j \neq \ell$ and the two rows are orthogonal, then

$$s_j + s_\ell \equiv 0 \pmod 4. \tag{6}$$

Indeed, if a,b,c,d count the four sign patterns of the two rows, then orthogonality gives $a+d = m/2$, which is even, and $s_j + s_\ell = 2(a-d)$ is divisible by four. Let $T = \{j : c_j \text{ is odd}\}$. Since $\sum_j c_j$ is even, $|T|$ is even. Terms with even c_j vanish modulo four, while for $j \in T$ one has $c_j s_j \equiv s_j \pmod 4$. Pair the indices in T and apply (6). Hence the product of the signs is 1. □

Put $\rho_k = k^{-1/2}$. By Lemma 4.1, the closed balls $B(\omega,\rho_k)$ are pairwise disjoint. Define

$$\mathrm{Near}_k := \bigcup_{\omega\in\Lambda_k} B(\omega,\rho_k), \qquad \mathrm{Far}_k := \mathbb{T}^k \setminus \mathrm{Near}_k.$$

For a k-row partial Hadamard matrix set

$$I(H) := \int_{\|\beta\|\le\rho_k} \Phi_H(\beta)\,d\beta, \qquad J(H) := \int_{\mathrm{Far}_k} \Phi_H(\theta)\,d\theta.$$

Then Lemmas 3.1 and 4.2 give

$$X(H) = 2^m(2\pi)^{-k}\big[2^{2k-1}I(H) + J(H)\big]. \tag{7}$$

5. The central region

Throughout this section H is a k-row partial Hadamard matrix and $y_C = \beta \cdot H_C$.

Lemma 5.1 (Exact orthogonality identities)**.** *For every $\beta \in \mathbb{R}^k$,*

$$\sum_{C=1}^m (\beta\cdot H_C)^2 = m\|\beta\|^2, \qquad \|H_C\|^2 = k, \qquad \sum_{C,C'=1}^m (H_C \cdot H_{C'})^2 = m^2 k.$$

Proof. The partial Hadamard condition is exactly $HH^\mathsf{T} = mI_k$. The first identity is $\beta^\mathsf{T} HH^\mathsf{T}\beta = m\|\beta\|^2$, and the second is immediate. For the third, $\sum_{C,C'}(H_C\cdot H_{C'})^2 = \mathrm{tr}((HH^\mathsf{T})^2) = m^2k$. □

Lemma 5.2 (Taylor bound)**.** *For $|y| \le 1$,*

$$-\log\cos y = \frac{y^2}{2} + \frac{y^4}{12} + s(y), \qquad 0 \le s(y) \le \frac{y^6}{30}.$$

Proof. The power series of $-\log\cos y$ on $|y|<\pi/2$ has positive even coefficients, beginning with $y^2/2+y^4/12$. Hence $s(y)/y^6$ is nondecreasing in $|y|$ on $[0,1]$. The numerical inequality $-\log(\cos 1)-\frac{7}{12}<\frac{1}{30}$ then gives the claim. This inequality may be verified, for example, by alternating-series bounds for $\cos 1$ and a positive-term truncation of the exponential series. $\square$

Let γ denote the Gaussian law $N(0,I_k/m)$ on $\mathbb{R}^k$, and define $Q(\beta):=\sum_{C=1}^m y_C^4$ and $Y_6(\beta):=\sum_{C=1}^m y_C^6$.

Lemma 5.3 (Gaussian moments)**.** *One has*

$$\mathbb{E}_\gamma Q=\frac{3k^2}{m},\qquad \mathrm{Var}_\gamma(Q)\le\frac{96k^3}{m^2},\qquad \mathbb{E}_\gamma Y_6=\frac{15k^3}{m^2}.$$

Proof. The vector $(y_1,\dots,y_m)$ is centered Gaussian with $\mathrm{Var}(y_C)=k/m$ and $\mathrm{Cov}(y_C,y_{C'})=(H_C\cdot H_{C'})/m$. The first and third identities follow from the fourth and sixth moments of a centered Gaussian. For the variance, if (Y,Z) is centered jointly Gaussian with variances a,b and covariance r, Wick's formula gives $\mathbb{E}[Y^4Z^4]=9a^2b^2+72abr^2+24r^4$. Summing this identity and using Lemma 5.1 yields

$$\mathbb{E}Q^2\le\frac{9k^4}{m^2}+\frac{72k^3}{m^2}+\frac{24k^3}{m^2}.$$

Subtracting $(\mathbb{E}Q)^2=9k^4/m^2$ proves the variance bound. $\square$

Lemma 5.4 (Chi-square tail)**.** *If $Z\sim\chi_k^2$ and $\lambda\ge1$, then $\mathbb{P}(Z\ge\lambda k)\le\exp\bigl(-\frac{k}{2}(\lambda-1-\log\lambda)\bigr)$. If $\lambda\ge6$, then $\mathbb{P}(Z\ge\lambda k)\le e^{-k\lambda/4}$.*

Proof. Apply the exponential Markov inequality and optimize the moment generating function $(1-2s)^{-k/2}$ at $s=(1-1/\lambda)/2$. The second claim follows from $\lambda-1-\log\lambda\ge\lambda/2$ for $\lambda\ge6$. $\square$

Proposition 5.5 (Uniform central contribution)**.** *Assume $m\ge6k^2$. For every k-row partial Hadamard matrix H,*

$$I(H)=\Bigl(\frac{2\pi}{m}\Bigr)^{k/2}e^{-k^2/(4m)}(1+\eta_H),\qquad |\eta_H|\le\frac{k^3}{m^2}+2e^{-m/(4k)}.$$

Proof. On $A=\{\|\beta\|\le k^{-1/2}\}$, Cauchy–Schwarz gives $|y_C|\le1$. By Lemmas 5.1 and 5.2,

$$\Phi_H(\beta)=\exp\Bigl(-\frac{m\|\beta\|^2}{2}-\frac{Q(\beta)}{12}-S_6(\beta)\Bigr),\qquad 0\le S_6(\beta)\le\frac{Y_6(\beta)}{30}.$$

Therefore $I(H)=(2\pi/m)^{k/2}\mathbb{E}_\gamma[\mathbf{1}_Ae^{-Q/12-S_6}]$. Put $\mu=\mathbb{E}Q/12=k^2/(4m)\le1/24$ and $W=Q-\mathbb{E}Q$. Since $W/12\ge-\mu$, Taylor's theorem gives

$$\mathbb{E}e^{-W/12}\le1+e^\mu\frac{\mathrm{Var}(Q)}{288}\le1+0.35\frac{k^3}{m^2}.$$

This yields the required upper bound. For the lower bound, Jensen's inequality gives $\mathbb{E}e^{-Q/12}\ge e^{-\mu}$. Also $\mathbb{E}[\mathbf{1}_AS_6]\le\frac{1}{30}\mathbb{E}Y_6=k^3/(2m^2)$. Finally $m\|\beta\|^2\sim\chi_k^2$, and $m/k^2\ge6$, so Lemma 5.4 gives $\mathbb{P}(A^c)\le e^{-m/(4k)}$. Combining these estimates and using $e^{1/24}<1.043$ gives

$$\mathbb{E}[\mathbf{1}_Ae^{-Q/12-S_6}]\ge e^{-\mu}\Bigl(1-0.53\frac{k^3}{m^2}-1.05e^{-m/(4k)}\Bigr).$$

The stated symmetric bound follows after harmless rounding. $\square$

6. The far field

The estimates in this section apply first to arbitrary sign matrices, without orthogonality assumptions. For $0 < \tau \le 2$, define

$$g_\tau(y) := \min\{1 - \cos(2y), \tau\}, \qquad G_\tau(\theta, H) := \sum_{C=1}^{m} g_\tau(\theta \cdot H_C).$$

Lemma 6.1 (Product bound)**.** *For every $\theta \in \mathbb{T}^k$, $|\Phi_H(\theta)| \le \exp\bigl(-\tfrac{1}{4} G_\tau(\theta, H)\bigr)$.*

Proof. Since $\cos^2 z = 1 - \frac{1-\cos 2z}{2} \le 1 - \frac{g_\tau(z)}{2}$, we have $\Phi_H(\theta)^2 \le \prod_C \bigl(1 - \frac{g_\tau(\theta \cdot H_C)}{2}\bigr) \le e^{-G_\tau(\theta,H)/2}$. Take square roots. $\square$

For $\theta \in \mathbb{T}^k$, let

$$\mu(\theta) := 1 - \prod_{j=1}^{k} \cos(2\theta_j), \qquad \sigma(\theta)^2 := \sum_{j=1}^{k} \operatorname{dist}(\theta_j, (\pi/2)\mathbb{Z})^2, \qquad r(\theta) := \operatorname{dist}(\theta, \Lambda_k).$$

Lemma 6.2 (Geometry of the mean)**.** *Let v be uniform on $\{\pm 1\}^k$. Then*

1. *$\mathbb{E}_v[1 - \cos(2\theta \cdot v)] = \mu(\theta)$;*
2. *$\mu(\theta) \ge \min\{1/2, r(\theta)^2\}$;*
3. *if $\mu(\theta) < 1/2$, the nearest point of the full grid $(\pi/2)\mathbb{Z}^k$ is unique, belongs to Λ_k, and has distance $\sigma(\theta) < 1/\sqrt{2}$.*

Proof. The first identity follows from $\mathbb{E}_v e^{2i\theta \cdot v} = \prod_j \cos(2\theta_j)$. Choose a nearest grid point $\omega = (\pi/2)c$ and write $\theta = \omega + \beta$ with $|\beta_j| \le \pi/4$, so that $\sigma(\theta) = \|\beta\|$. If $\sum_j c_j$ is odd, then $\prod_j \cos(2\theta_j) \le 0$ and $\mu(\theta) \ge 1 \ge \min\{1/2, r(\theta)^2\}$.

Suppose instead that $\sum_j c_j$ is even, so that $\omega \in \Lambda_k$. We claim $\sigma(\theta) = r(\theta)$. Indeed $\Lambda_k \subset (\pi/2)\mathbb{Z}^k$, and the distance to a subset is at least the distance to the ambient set, so $\sigma(\theta) \le r(\theta)$; conversely $\omega \in \Lambda_k$ gives $r(\theta) \le \|\theta - \omega\| = \sigma(\theta)$. (The inclusion alone would give the inequality in the wrong direction, which is why the parity case matters here.) Now $\cos(2\beta_j) \le e^{-2\beta_j^2}$ for $|\beta_j| \le \pi/4$, so $\prod_j \cos(2\beta_j) \le e^{-2\sigma(\theta)^2}$ and

$$\mu(\theta) \ge 1 - e^{-2\sigma(\theta)^2} \ge \min\{1/2, \sigma(\theta)^2\} = \min\{1/2, r(\theta)^2\},$$

where the middle step holds because $1 - e^{-2s} \ge s$ on $[0, 1/2]$ and $1 - e^{-2s} \ge 1 - e^{-1} > 1/2$ for $s \ge 1/2$.

If $\mu < 1/2$, no coordinate can lie halfway between two grid points, since that would give $\cos(2\theta_j) = 0$ and hence $\mu = 1$; so the nearest grid point is unique. The odd-parity case is impossible, and the preceding bound forces $\sigma^2 < 1/2$. $\square$

Lemma 6.3 (A truncated mean on dyadic shells)**.** *Suppose $\theta = \omega + \beta$, where $\omega \in \Lambda_k$, $|\beta_j| \le \pi/4$, and $\sigma = \|\beta\|$. Let $p \ge 1$ satisfy $2^{-p-1} < \sigma^2 \le 2^{-p} \le \frac{1}{2}$, and put $\tau_p = \min\{2, 48 \cdot 2^{-p}\}$. Then*

$$W_p(\theta) := \mathbb{E}_v g_{\tau_p}(\theta \cdot v) \ge \tfrac{3}{8} 2^{-p}, \qquad \frac{W_p(\theta)}{\tau_p} \ge \frac{1}{128}.$$

Proof. Set $A(v) = 1 - \cos(2\theta \cdot v)$. Since $\sum_j c_j$ is even and $v \in \{\pm 1\}^k$, we have $c \cdot v \equiv \sum_j c_j \equiv 0 \pmod 2$, so $\omega \cdot v \in \pi\mathbb{Z}$ and $A(v) = 1 - \cos(2\beta \cdot v)$: translation by $\omega \in \Lambda_k$ does not change A. By Lemma 6.2 and $\sigma^2 \le 1/2$, $\mathbb{E}A \ge \sigma^2$. Moreover $1 - \cos 2x = 2\sin^2 x \le 2x^2$ gives $A \le 2(\beta \cdot v)^2$, and the fourth moment of a Rademacher sum, $\mathbb{E}(\beta \cdot v)^4 \le 3\sigma^4$, gives $\mathbb{E}A^2 \le 4\mathbb{E}(\beta\cdot v)^4 \le 12\sigma^4$. If $p \le 4$, then $48\cdot 2^{-p} \ge 3 > 2$, so $\tau_p = 2$, the truncation is inactive, and the conclusion follows from $W_p = \mathbb{E}A > 2^{-p-1} \ge \frac{3}{8}2^{-p}$. If $p \ge 5$, then using $\sigma^2 \le 2^{-p}$ and $\mathbb{E}A \ge \sigma^2$,

$$W_p \ge \mathbb{E}A - \mathbb{E}[A\mathbf{1}_{\{A>\tau_p\}}] \ge \mathbb{E}A - \frac{\mathbb{E}A^2}{\tau_p} \ge \mathbb{E}A - \frac{12\sigma^4}{48\cdot 2^{-p}} \ge \mathbb{E}A - \frac{\sigma^2}{4} \ge \frac{3}{4}\mathbb{E}A > \frac{3}{8}2^{-p}.$$

Dividing by τ_p gives the second bound, since $\frac{3}{8}/48 = \frac{1}{128}$. $\square$

Lemma 6.4 (Lipschitz estimate)**.** *For all $\theta, \theta' \in \mathbb{T}^k$, $|G_\tau(\theta,H) - G_\tau(\theta',H)| \le 2m\sqrt{k}\,\|\theta - \theta'\|$.*

Proof. The function g_τ is 2-Lipschitz, while $|\langle \theta-\theta', H_C\rangle| \le \sqrt{k}\|\theta-\theta'\|$. Sum over C. $\square$

6.1. **Shells and nets.** Put $P := \max\{0, \lceil \log_2 k\rceil - 1\}$. Decompose Far_k into

$$F_A := \{\theta \in \mathrm{Far}_k : \mu(\theta) \ge 1/2\},$$
$$F_p := \{\theta \in \mathrm{Far}_k : \mu(\theta) < 1/2,\ 2^{-p-1} < \sigma(\theta)^2 \le 2^{-p}\}, \qquad 1 \le p \le P.$$

By Lemma 6.2, these sets cover Far_k. Set $\eta_A = \frac{1}{16\sqrt{k}}$ and $\eta_p = \frac{3\cdot 2^{-p}}{64\sqrt{k}}$, and choose nets $\mathcal{N}_A \subset F_A$ and $\mathcal{N}_p \subset F_p$ that are maximal subject to having all pairwise distances *greater than* the corresponding scale. Maximality then gives covering radius at most that scale, which is what Proposition 6.9 uses.

Lemma 6.5 (Net cardinalities)**.** *For all relevant p, $|\mathcal{N}_A| \le e^{k\log(200k^2)}$ and $|\mathcal{N}_p| \le e^{k\log(200k^2)}$.*

Proof. Partition $\mathbb{T}^k$ into axis-parallel cells of side h. A cell has diameter $h\sqrt{k}$, so a set with pairwise distances *greater than* η has at most one point in each cell when $h = \eta/\sqrt{k}$. This gives $|\mathcal{N}_A| \le (1 + 2\pi\sqrt{k}/\eta_A)^k = (1+32\pi k)^k \le (102k)^k$, using $32\pi < 100.54$.

For F_p, every point lies in one of the 2^{2k-1} cubes $\omega + [-\pi/4, \pi/4]^k$, $\omega \in \Lambda_k$, by Lemma 6.2(c). An interval of length $\pi/2$ on the circle intersects at most $\pi/(2h) + 2$ cells of side h. Here $h = \eta_p/\sqrt{k}$ and, whenever F_p is present, $k \ge 3$ and $2^p \le k$. Hence

$$\frac{\pi}{2h} + 2 = \frac{32\pi}{3}k2^p + 2 \le 34k^2,$$

since $32\pi/3 < 33.52$ and $0.48k^2 \ge 2$ for $k \ge 3$. Consequently $|\mathcal{N}_p| \le 2^{2k-1}(34k^2)^k \le (136k^2)^k \le (200k^2)^k$. This corrected endpoint count is the reason for the harmless numerical constant 200. $\square$

Lemma 6.6 (Bernstein-type lower tail)**.** *Let $X_1, \dots, X_m$ be i.i.d. random variables in $[0,\tau]$ with mean $W > 0$. Then*

$$\mathbb{P}\Big(\sum_{C=1}^{m} X_C \le \frac{mW}{2}\Big) \le \exp\Big(-\frac{mW}{8\tau}\Big).$$

Proof. Take $\lambda = 1/(2\tau)$. Since $e^{-z} \le 1 - z + z^2/2$ for $z \ge 0$ and $X_C^2 \le \tau X_C$, $\mathbb{E}e^{-\lambda X_C} \le 1 - \lambda W + \frac{\lambda^2\tau W}{2} \le e^{-3\lambda W/4}$. Markov's inequality applied to $e^{-\lambda\sum X_C}$ gives the result. $\square$

A uniformly random $k\times m$ sign matrix has independent uniform columns in $\{\pm1\}^k$.

Definition 6.7 (Good matrices)**.** A $k\times m$ sign matrix H is *good* if

$$G_2(\theta', H) \ge m/4 \quad \text{for every } \theta' \in \mathcal{N}_A, \qquad G_{\tau_p}(\theta', H) \ge (3/16)m2^{-p} \quad \text{for every } \theta' \in \mathcal{N}_p,\ 1 \le p \le P.$$

Lemma 6.8 (Good matrices are typical)**.** *For a uniform random $k \times m$ sign matrix,*

$$\mathbb{P}(H \text{ is not good}) \le \exp\Big(k\log(200k^2) + \log(P+1) - \frac{m}{1024}\Big).$$

Proof. At a point of $\mathcal{N}_A \subset F_A$, the truncation at $\tau = 2$ is inactive, so $W = \mu(\theta') \ge 1/2$ by the definition of F_A; apply Lemma 6.6 to get failure probability at most $e^{-m/32}$. At a point of $\mathcal{N}_p$, Lemmas 6.3 and 6.6 give failure probability at most $e^{-m/1024}$. A union bound over the at most $P+1$ nets, each of size at most $e^{k\log(200k^2)}$ by Lemma 6.5, with the weaker per-point bound $e^{-m/1024}$, proves the claim. For $k \le 2$ one has $P = 0$ and $\log(P+1) = 0$; the bound remains valid, and in that range $\mathrm{Far}_k = F_A$, since a point with $\mu < 1/2$ would need $\sigma^2 = r^2 > 1/k \ge 1/2$ while $\mu < 1/2$ forces $\sigma^2 < 1/2$. □

Proposition 6.9 (Pointwise far-field decay)**.** *If H is good, then*

$$|\Phi_H(\theta)| \le \exp\Big(-\frac{3m}{128}\sigma(\theta)^2\Big), \quad \theta \in F_p, \qquad |\Phi_H(\theta)| \le e^{-m/32}, \quad \theta \in F_A.$$

Proof. For $\theta \in F_p$, choose $\theta' \in \mathcal{N}_p$ with $\|\theta - \theta'\| \le \eta_p$. By Definition 6.7 and Lemma 6.4,

$$G_{\tau_p}(\theta, H) \ge \frac{3m}{16}2^{-p} - 2m\sqrt{k}\eta_p = \frac{3m}{32}2^{-p}.$$

Apply Lemma 6.1 and use $\sigma(\theta)^2 \le 2^{-p}$. The argument on F_A is identical: $G_2(\theta, H) \ge m/4 - 2m\sqrt{k}\eta_A = m/8$. □

Corollary 6.10 (Far integral)**.** *If H is good and $m \ge 128k^2$, then*

$$|J(H)| \le 2^{2k-1}\Big(\frac{2\pi}{m}\Big)^{k/2}\exp\Big(1.54k - \frac{3m}{256k}\Big) + (2\pi)^k e^{-m/32}.$$

Proof. The contribution of F_A is at most $(2\pi)^k e^{-m/32}$. On the union of the dyadic shells, use the unique nearest $\omega \in \Lambda_k$ and Proposition 6.9 to obtain

$$\sum_p \int_{F_p} |\Phi_H| \le |\Lambda_k| \int_{\|\beta\| > k^{-1/2}} e^{-3m\|\beta\|^2/128}\, d\beta.$$

The integral is the normalizing constant for a Gaussian with covariance $64I_k/(3m)$, multiplied by the tail probability $\mathbb{P}(\chi_k^2 > 3m/(64k))$. Writing the threshold as λk gives $\lambda = 3m/(64k^2) \ge 6$. Apply Lemma 5.4. Finally $(64/3)^{k/2} \le e^{1.54k}$. □

Remark 6.11 (Why an exceptional set is necessary)*.* There is no uniform estimate $|\Phi_H(\theta)| \le e^{-cm}$ on all of Far_k for every partial Hadamard matrix. Take $k = 4$, $8 \mid m$, and use each of the eight vectors $v \in \{\pm 1\}^4$ with $v_1v_2v_3v_4 = 1$ exactly $m/8$ times as a column. The resulting matrix is a 4-row partial Hadamard matrix. At $\theta^* = (\pi/4)(1,1,1,1)$, every column satisfies $\theta^* \cdot v \in \pi\mathbb{Z}$, so $|\Phi_H(\theta^*)| = 1$. Yet $\mathrm{dist}(\theta^*, \Lambda_4) = \pi/2 > 1/2 = \rho_4$, so $\theta^* \in \mathrm{Far}_4$.

7. One inductive step

Fix $n \ge 2$ and $1 \le k \le n-1$. Assume

$$m \ge 20000\, n^2 L. \tag{8}$$

Then $m \ge 64000n^2$, so all hypotheses of the central and far-field estimates hold. Define

$$V_k := 2^m 2^{2k-1}(2\pi m)^{-k/2} e^{-k^2/(4m)}. \tag{9}$$

Since $2\pi m > 16$, one has $0 < V_k \le 2^m$.

Proposition 7.1 (Extension count for a good partial Hadamard matrix). *If H is a good k-row partial Hadamard matrix, then $\left|\frac{X(H)}{V_k} - 1\right| \le \delta_k$, where*

$$\delta_k := \frac{k^3}{m^2} + 2e^{-m/(4k)} + 1.01e^{1.54k-3m/(256k)} + 2.02e^{(k/2)L-m/32}.$$

Proof. Insert Proposition 5.5 and Corollary 6.10 into (7). The central main term is exactly V_k. After division by V_k, the first far-field term is multiplied by $e^{k^2/(4m)} \le 1.01$. The second is bounded by $2e^{kL/2}e^{k^2/(4m)}e^{-m/32} \le 2.02e^{kL/2-m/32}$. $\square$

Set $M_j := 2^m \prod_{i=1}^{j-1} V_i$, with $M_1 = 2^m$. Also define

$$\psi_k := 2\exp\Big(2n^2L + 2n\log(200n^2) - \frac{m}{1024}\Big), \qquad e_k := \delta_k + \psi_k.$$

Proposition 7.2 (One-step induction). *Assume (8) and $N_{k,m} \ge M_k/2$. Then*

$$\left|\frac{N_{k+1,m}}{N_{k,m}V_k} - 1\right| \le e_k.$$

Proof. Let B_k be the number of bad k-row partial Hadamard matrices. By Lemma 6.8, the number of bad $k\times m$ sign matrices is at most $2^{km}q_k$, where $q_k := \exp\big(2k\log(200k^2) - \frac{m}{1024}\big)$, so $B_k \le 2^{km}q_k$. From (5), Proposition 7.1, and $X(H), V_k \le 2^m$,

$$\left|\frac{N_{k+1,m}}{N_{k,m}V_k} - 1\right| \le \delta_k + \frac{B_k 2^m}{N_{k,m}V_k}.$$

It remains to bound the exceptional contribution. Put $\Pi_k := M_k/2^{km} = \prod_{i=1}^{k-1} V_i/2^m$. Since $V_i/2^m = 2^{2i-1}(2\pi m)^{-i/2}e^{-i^2/(4m)}$, we obtain

$$\Pi_k \ge \exp\Big(-\frac{k^2L}{4} - \frac{k^3}{12m}\Big) \ge e^{-0.28k^2L}.$$

Similarly $V_k/2^m \ge e^{-kL}$. Therefore

$$\frac{N_{k,m}V_k}{2^{(k+1)m}} \ge \frac{1}{2}e^{-0.28k^2L-kL} \ge \frac{1}{2}e^{-2k^2L}.$$

Consequently

$$\frac{B_k 2^m}{N_{k,m}V_k} \le 2\exp\Big(2k^2L + 2k\log(200k^2) - \frac{m}{1024}\Big) \le \psi_k. \qquad \square$$

Lemma 7.3 (Summed one-step error). *Under (8), $\sum_{k=1}^{n-1} e_k \le E(n,t)$.*

Proof. Each family of summands is increasing in k. Thus

$$\sum_{k<n} \frac{k^3}{m^2} \le \frac{n^4}{4m^2}, \qquad \sum_{k<n} 2e^{-m/(4k)} \le 2ne^{-m/(4n)},$$

$$\sum_{k<n} 1.01e^{1.54k-3m/(256k)} \le 2ne^{2n-m/(86n)}, \qquad \sum_{k<n} 2.02e^{(k/2)L-m/32} \le 3ne^{(n/2)L-m/32},$$

$$\sum_{k<n} \psi_k \le 2n\exp\Big(2n^2L + 2n\log(200n^2) - \frac{m}{1024}\Big).$$

Adding the bounds gives (2). $\square$

8. Elementary parameter bounds

Put $\Theta := \frac{m}{n^2L}$. Condition (8) is $\Theta \geq 20000$.

Lemma 8.1 (Control of E)**.** *If $\Theta \geq 20000$, then $E(n,t) \leq F(\Theta)$, where*

$$F(x) := \frac{1}{x} + 2e^{-0.4x} + 82e^{-0.009x} + 23e^{-0.05x} + 471e^{-0.0015x}.$$

The function F is decreasing, tends to zero, and satisfies $F(20000) < 10^{-4} < \frac{1}{2}$.

Proof. Use $m = n^2L\Theta$, $L > 3.2$, and the elementary estimate

$$xe^{-axy} \leq \frac{2}{ea}e^{-ay/2}, \qquad x, y \geq 1, \tag{10}$$

which follows from $xy \geq (x+y)/2$ and optimization in x. The five terms of E are bounded respectively by $\frac{1}{\Theta}$, $2e^{-0.4\Theta}$, $82e^{-0.009\Theta}$, $23e^{-0.05\Theta}$, $471e^{-0.0015\Theta}$. For the final term, note that under $\Theta \geq 20000$, $\log(200n^2) \leq 3.66L$ and hence $2n^2L + 2n\log(200n^2) - m/1024 \leq -m/2048$. The remaining estimates are direct substitutions into (10). Monotonicity is immediate, and the numerical value at 20000 is less than 5.001×10^{-5}. ☐

Lemma 8.2 (Control of the safety term)**.** *If $\Theta \geq 40000$, then $S(n,t) \leq G(\Theta)$, where $G(x) := \left(\frac{20000}{x}\right)^4(13 + 4\log x)$. The function G is decreasing on $[40000, \infty)$ and tends to zero.*

Proof. Let $x = 20000/\Theta \leq 1/2$. The prefactor in (3) is at most $n^2L/2$. Since $L = \log(2\pi) + 2\log n + \log L + \log\Theta \leq 3.68 + 4\log n + 2\log\Theta$, we get

$$S(n,t) \leq \frac{n^2}{2}(3.68 + 4\log n)x^{n^2} + n^2(\log\Theta)x^{n^2}.$$

Because $x \leq 1/2$ and $n^2 \geq 4$, $x^{n^2} \leq 2^{4-n^2}x^4$. Elementary maximization over integers $n \geq 2$ gives $n^2(3.68 + 4\log n)2^{4-n^2} < 25.89$ and $n^2 2^{4-n^2} \leq 4$. Thus $S \leq x^4(13 + 4\log\Theta)$. Differentiation proves monotonicity. ☐

9. Proof of the explicit theorem

Proof of Theorem 2.1. First compute the product of the row factors. Using

$$\sum_{k=1}^{n-1}(2k-1) = (n-1)^2, \qquad \sum_{k=1}^{n-1} k = \frac{n(n-1)}{2}, \qquad \sum_{k=1}^{n-1} k^2 = \frac{n(n-1)(2n-1)}{6},$$

we find

$$M_n = 2^m\prod_{k=1}^{n-1} V_k = 2^{mn+(n-1)^2}(2\pi m)^{-n(n-1)/4}\exp\Big(-\frac{n(n-1)(2n-1)}{24m}\Big) = B_{n,4t}.$$

Suppose first that (8) fails. Then $20000n^2L/m > 1$, and hence $S(n,t) > \frac{n^2L}{4} + \frac{n^3}{12m}$. The trivial bound $N_{n,m} \leq 2^{nm}$ gives

$$\frac{N_{n,m}}{B_{n,m}} \leq 2^{-(n-1)^2}(2\pi m)^{n(n-1)/4}\exp\Big(\frac{n(n-1)(2n-1)}{24m}\Big) \leq \exp\Big(\frac{n^2L}{4} + \frac{n^3}{12m}\Big) < e^{S(n,t)} \leq 1 + \varepsilon(n,t).$$

On the other side, $N_{n,m}/B_{n,m} \geq 0$, so $1 - N_{n,m}/B_{n,m} \leq 1$. Since $S(n,t) > 3.2$, $\varepsilon(n,t) > e^{3.2} - 1 > 1$. The theorem follows in this case.

Now assume (8). By Lemma 8.1, $E(n,t) < 1/2$. We prove inductively that

$$N_{k,m} = M_k \prod_{j=1}^{k-1}(1+\xi_j), \qquad |\xi_j| \le e_j, \qquad \prod_{j=1}^{k-1}(1+\xi_j) \ge \frac{1}{2}.$$

The statement is trivial at $k=1$. If it holds at k, then $N_{k,m} \ge M_k/2$, so Proposition 7.2 defines ξ_k with $|\xi_k| \le e_k$. By Lemma 7.3, $\sum_{j=1}^{n-1} e_j \le E(n,t) \le \frac{1}{2}$. Therefore

$$\prod_{j=1}^{k}(1+\xi_j) \ge \prod_{j=1}^{k}(1-e_j) \ge 1 - \sum_{j=1}^{k} e_j \ge \frac{1}{2}.$$

The induction closes. Finally,

$$\left|\frac{N_{n,m}}{M_n} - 1\right| = \left|\prod_{j=1}^{n-1}(1+\xi_j) - 1\right| \le e^{\sum_j |\xi_j|} - 1 \le e^{E(n,t)} - 1 \le \varepsilon(n,t).$$

Since $M_n = B_{n,4t}$, the proof is complete. □

10. Asymptotic consequences

Proof of Theorem 2.2. If $\Theta = m/(n^2L) \to \infty$, then eventually $\Theta \ge 40000$. By Lemmas 8.1 and 8.2, $E(n,t) + S(n,t) \le F(\Theta) + G(\Theta) \to 0$. Thus Theorem 2.1 gives the claimed uniform convergence. □

Lemma 10.1 (Power laws imply the natural regime)**.** *Let $0 < \delta \le 1/2$, put $u = 2 + 2\delta$ and $K = 6 + 2u/\delta$. If $\omega := t/n^u \ge 1$ and $n \ge 2$, then*

$$\frac{m}{n^2 \log(2\pi m)} \ge \frac{2}{K}\omega^{1/2}.$$

Proof. Since $m = 4n^u\omega$, $L = \log(8\pi n^u \omega) = \log(8\pi) + u\log n + \log\omega$. Using $\log x \le x^a/a$ for $x \ge 1$, first with $a = \delta/2$ and then with $a = 1/2$, gives $\log n \le (2/\delta)n^{\delta/2}$ and $\log\omega \le 2\omega^{1/2}$. Since $\log(8\pi) \le 6$ and $n^{\delta/2}, \omega^{1/2} \ge 1$, we get $L \le K(n^{\delta/2} + \omega^{1/2})$. Therefore, writing $n^u = n^2 n^{2\delta}$,

$$\frac{m}{n^2 L} \ge \frac{4n^{2\delta}\omega}{K(n^{\delta/2} + \omega^{1/2})}.$$

Finally $n^{\delta/2} \le n^{2\delta}$ and $\omega^{1/2} \le \omega$, so $\omega^{1/2}(n^{\delta/2} + \omega^{1/2}) \le 2n^{2\delta}\omega$, which is the claim. □

Proof of Corollary 2.3. Given an exponent $u > 2$, put $\delta = \min\{(u-2)/2, 1/2\}$ and $u_0 = 2 + 2\delta \le u$. The assumption $t/n^u \to \infty$ implies $t/n^{u_0} \to \infty$. Hence Lemma 10.1 shows that $m/(n^2L) \to \infty$. Apply Theorem 2.2. □

Lemma 10.1 is sharp, in the following precise sense.

Lemma 10.2 (No exponent at most two suffices)**.** *Let $1 \le u \le 2$. For all $N \ge 2$ and $\omega \ge 1$, writing $m = 4N^u\omega$,*

$$\frac{m}{N^2 \log(2\pi m)} \le \frac{4\omega}{\log N}. \tag{11}$$

Consequently there is a family with $\omega \to \infty$ along which the left side stays bounded by $1/2$: take $\omega = j$ and $N = e^{8j}$, $j \ge 1$.

Proof. Since $u \le 2$ and $N \ge 2$, we have $N^u \le N^2$; and since $u \ge 1$, $\omega \ge 1$, $\log(2\pi m) = \log(8\pi) + u \log N + \log \omega \ge \log N > 0$. Hence

$$\frac{m}{N^2 \log(2\pi m)} = \frac{4N^u\omega}{N^2 \log(2\pi m)} \le \frac{4N^2\omega}{N^2 \log N} = \frac{4\omega}{\log N}.$$

For the family, $\log N = 8j = 8\omega$, so the bound is $4\omega/(8\omega) = 1/2$. □

The witness can be taken to consist of genuine *integer* pairs, which is the domain of the counting problem; the relaxation to real parameters above is a convenience, not a load-bearing step.

Lemma 10.3 (Integer witness)**.** *Let $1 \le u \le 2$ and let $j \ge 1$ be an integer. Put*

$$n_j := \lceil e^{16j} \rceil, \qquad t_j := \lceil j\, n_j^u \rceil, \qquad m_j := 4t_j.$$

Then $n_j \ge 2$, $t_j \ge 1$, $t_j/n_j^u \ge j$, and

$$\frac{m_j}{n_j^2 \log(2\pi m_j)} \le \frac{1}{2}.$$

Proof. Clearly $n_j \ge 2$ and $t_j \ge 1$, and $t_j \ge j\, n_j^u$ gives $t_j/n_j^u \ge j$, which tends to infinity. Ceilings cost at most one, so $t_j \le j\, n_j^u + 1$; since $u \le 2$ and $4 \le 4j\, n_j^2$,

$$m_j = 4t_j \le 4j\, n_j^u + 4 \le 4j\, n_j^2 + 4j\, n_j^2 = 8j\, n_j^2.$$

Since $u \ge 1$ and $j \ge 1$ we have $n_j \le n_j^u \le j\, n_j^u \le t_j$, whence $2\pi m_j \ge n_j$ and $\log(2\pi m_j) \ge \log n_j \ge 16j$. Therefore

$$\frac{m_j}{n_j^2 \log(2\pi m_j)} \le \frac{8j\, n_j^2}{n_j^2 \cdot 16j} = \frac{1}{2}.$$ □

The family of Lemma 10.2 satisfies the power law with exponent u with room to spare, $t/N^u = \omega = j \to \infty$, yet the near-quadratic quantity never exceeds $1/2$. So no exponent $u \le 2$ forces the hypothesis of Theorem 2.2, while by Lemma 10.1 every $u > 2$ does. The infimum of the exponents reachable by this argument is therefore exactly 2, and it is not attained. Both directions are formally verified; in the accompanying Lean development the positive direction is `bridges_of_two_lt`, the negative direction `not_forces_of_le_two` with its integer form `exists_int_witness_of_le_two`, and the conclusion `sInf_admissibleExponents`.

A remark on what "forces" must mean here. The relevant notion is the one used in Corollary 2.3: along every sequence of pairs (n_i, t_i) with $\omega_i = t_i/n_i^u \to \infty$, and with n_i free to vary, the quantity $m_i/(n_i^2 \log 2\pi m_i)$ tends to infinity. The uniformity in n is essential, not a technical convenience: if instead n were held fixed while $\omega \to \infty$, then $m/(n^2 \log 2\pi m)$ would diverge like $\omega/\log\omega$ for *every* exponent u, so every $\alpha \ge 1$ would count as admissible and the constant would carry no information. In the Lean development this equivalence is `forces_iff_sequential`, which shows the quantifier form used there agrees with the sequential premise above.

We stress what the negative direction does and does not say. It shows that the *implication* "power law with exponent $u \le 2 \Rightarrow$ near-quadratic hypothesis" is false, which is precisely why the present method stops at 2. It does *not* show that the asymptotic formula itself fails for such u; whether it holds there is open, as recorded in Section 12.

11. Exact low-dimensional checks

The correction factor can be checked against exact formulas. For $m \in 4\mathbb{N}$,

$$N_{2,m} = 2^m \binom{m}{m/2}, \qquad N_{3,m} = 2^m \frac{m!}{((m/4)!)^4}, \qquad N_{4,m} = 2^m \sum_{a+b=m/4} \frac{m!}{(a!)^4 (b!)^4}.$$

For the last identity, normalize the first row to all $+1$. The eight column types of the remaining three rows have multiplicities determined by a single third-order Fourier coefficient: the four types with product $+1$ occur a times each and the four types with product -1 occur b times each.

Table 1. Relative error with and without the correction factor.

n	m	$N_{n,m}/B_{n,m} - 1$	$N_{n,m}/A_{n,m} - 1$
2	8	$+7.99 \times 10^{-5}$	-3.069×10^{-2}
2	32	$+1.27 \times 10^{-6}$	-7.781×10^{-3}
2	256	$+2.48 \times 10^{-9}$	-9.761×10^{-4}
3	8	$+1.30 \times 10^{-3}$	-1.435×10^{-1}
3	32	$+2.15 \times 10^{-5}$	-3.829×10^{-2}
3	256	$+4.22 \times 10^{-8}$	-4.871×10^{-3}
4	16	$+1.49 \times 10^{-3}$	-1.953×10^{-1}
4	32	-3.76×10^{-4}	-1.039×10^{-1}
4	40	-2.61×10^{-4}	-8.402×10^{-2}

These calculations are consistency checks, not inputs to the proof.

12. Scope and limitations

The local expansion in Proposition 5.5 has accumulated relative error of order n^4/m^2, which already tends to zero under $m/n^2 \to \infty$. The logarithmic loss comes entirely from comparing the exceptional set of far-field matrices, whose ambient density is $e^{-\Omega(m)}$, with the density of partial Hadamard matrices, which is roughly $e^{-\Theta(n^2 \log m)}$. Removing the logarithm would require new information about the far field under the partial-Hadamard conditioning; a universal pointwise estimate is ruled out by Remark 6.11.

The proof establishes a corrected asymptotic formula. It does not imply $N_{n,m} \sim A_{n,m}$ below the cubic scale, because $R_{n,m}$ can tend to zero. Nor does it prove the endpoint regime $m/n^2 \to \infty$. In particular, the upper bound 2 for the critical exponent is an infimum over admissible power laws and is not attained by the present argument.

13. Use of AI tools

The proof strategy and an initial detailed draft were developed with the assistance of Hyra, a research agent [4]. The present manuscript was subsequently subjected to an independent line-by-line mathematical audit, and local issues in the original draft were repaired, including the description of the universal saddle lattice and an endpoint term in the net-cardinality estimate. AI-generated judgments were not treated as certificates of correctness; the mathematical claims rest on the explicit argument above.

This disclosure follows the general principle used in other Hyra-assisted mathematical work [5].

Appendix A. Numerical constants

For convenience, we collect the nontrivial numerical inequalities used in the proof:

$$\log(8\pi) > 3.2, \qquad -\log(\cos 1) - \tfrac{7}{12} < \tfrac{1}{30}, \qquad e^{1/24} < 1.043, \qquad \sqrt{64/3} < e^{1.54},$$
$$\tfrac{3}{256} > \tfrac{1}{86}, \qquad \tfrac{1}{4} + \tfrac{1}{38} < 0.28, \qquad \log 200 < 1.66 \cdot 3.2, \qquad 2 \cdot 5.66 \cdot 1024 < 20000.$$

All remaining constants are exact rational quantities or are obtained by rounding in the safe direction.

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